

2010376103 CSE4261 Assignment-14

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Dataset and Preprocessing

For this assignment, a **subset of the ImageNet dataset** was used, containing **20 randomly selected classes**. Each image was resized to a fixed dimension of 128×128 **pixels** and normalized to have zero mean and unit variance across RGB channels. To ensure consistent experiments, the dataset was divided into training, validation, and test sets:

- **Training set:** 25,729 samples
- **Validation set:** 500 samples
- **Test set:** 500 samples

All experiments were performed using PyTorch with a batch size of 64 and reproducibility ensured by a fixed random seed (42). When applicable, standard image augmentation was applied (random horizontal flip, random rotation) to improve generalization.

1 Training a ViT-Based Classifier for 20 Classes of ImageNet

[Github code link](#)

1.1 Vision Transformer (ViT) Architecture

The Vision Transformer (ViT) divides each image into fixed-size patches, linearly embeds them, adds positional encodings, and passes the sequence through multiple Transformer encoder blocks. Each encoder block contains:

- Multi-head self-attention mechanism
- Layer normalization
- Feed-forward MLP block

The classification is done via a learnable [CLS] token whose output embedding passes through a final linear layer.

1.2 Training Configuration

The ViT model was trained on the ImageNet-20 dataset using the Adam optimizer and a cosine annealing learning rate scheduler. Training lasted for approximately **46.67 minutes**, achieving the following performance:

- **Best Validation Accuracy:** 60.20%
- **Final Test Accuracy:** 60.20%

These results serve as the baseline ViT performance for subsequent experiments.

2 Comparison with CNN and FCFNN

2.1 Fully Connected Feed-Forward Neural Network (FCFNN)

The FCFNN consists of three dense layers with ReLU activations, followed by dropout regularization and a softmax output layer. This architecture treats flattened image pixels as independent features and lacks spatial awareness.

- Training time: 14.34 minutes
- Validation accuracy: 22.00%
- Test accuracy: 18.60%
- Test loss: 2.7176

The poor performance is due to the inability to capture local spatial dependencies among neighboring pixels.

2.2 Convolutional Neural Network (CNN)

The CNN architecture included multiple convolutional and pooling layers for spatial feature extraction, followed by fully connected layers for classification.

- Training time: 25.39 minutes
- Validation accuracy: 76.80%
- Test accuracy: 75.00%
- Test loss: 0.7674

CNN outperformed both FCFNN and ViT on this small-scale dataset, showing that convolutional inductive biases (locality, weight sharing) are beneficial when training data is limited.

3 Effect of the Number of Attention Heads in ViT

To investigate how the number of attention heads influences performance, two ViT configurations were trained: 4 heads and 8 heads.

3.1 Experimental Summary

Heads	Best Val Acc (%)	Final Test Acc (%)	Train Time (min)
4	56.60	60.20	44.82
8	61.20	62.20	49.15

3.2 Observation

Increasing the number of attention heads improved both validation and test accuracy slightly but also increased training time. This suggests that multi-head attention helps the model capture diverse contextual relationships between patches, but the benefit saturates quickly for small datasets.

4 Effect of Positional Embedding

Three types of positional embeddings were tested using the 4-head ViT model:

Positional Embedding	Best Val Acc (%)	Final Test Acc (%)	Train Time (min)
Learnable	58.80	62.80	44.71
Sine	63.00	59.80	44.81
None	59.80	61.40	44.79

4.1 Observation

Sine-based embeddings achieved the best validation accuracy, while learnable embeddings gave slightly higher test accuracy. This indicates that the ViT can adaptively learn positional information when sufficient data is available, but fixed sinusoidal encodings may generalize better under limited data.

5 Discussion: Handling ViT vs. CNN and FCFNN

5.1 Challenges in ViT Training

- **Data Requirement:** ViT models typically require large-scale datasets to learn spatial priors, unlike CNNs that have built-in locality.
- **Computational Cost:** Self-attention scales quadratically with sequence length, making ViTs slower on high-resolution images.
- **Positional Encoding Sensitivity:** Improper positional encoding can degrade spatial understanding.

5.2 Comparative Insights

- FCFNNs underperform due to the absence of spatial structure.
- CNNs leverage local feature extraction, achieving high accuracy with less data.
- ViTs can rival or surpass CNNs only with more data and compute power, demonstrating their scalability rather than efficiency.

6 Conclusion

This assignment explored training and analyzing a Vision Transformer (ViT) on a 20-class ImageNet subset and comparing it against CNN and FCFNN baselines. ViT performance improved with more attention heads and suitable positional embeddings but remained below CNN accuracy due to limited data and compute. ViTs offer interpretability and scalability advantages, but they are computationally more expensive and harder to train in small-data regimes.